ADCs and 1-bit measurements, explained with pictures

How an ADC works, what the research papers say, why NOMP struggles with coarse measurements, and what the IABR-Net results show

How to read this document. Every section has a figure. Figures 4 to 7 are made from the project's own simulator (one real simulated scene: 16 x 16 beams, 3 paths, 15 dB SNR). Statements about papers come from the project's study notes, in which each paper was read in full unless marked otherwise. Result numbers are the project's verified results (1000 fresh scenes per case). Numbers marked "demo" are small illustrations computed for this document and are not paper results.

1. What is an ADC?

An antenna delivers an **analog** signal: a smooth voltage that can take any value. A computer can only store numbers. The **analog-to-digital converter (ADC)** does the conversion in two steps: it measures the signal at regular times (**sampling**), and it rounds each measurement to the nearest allowed level (**quantisation**).

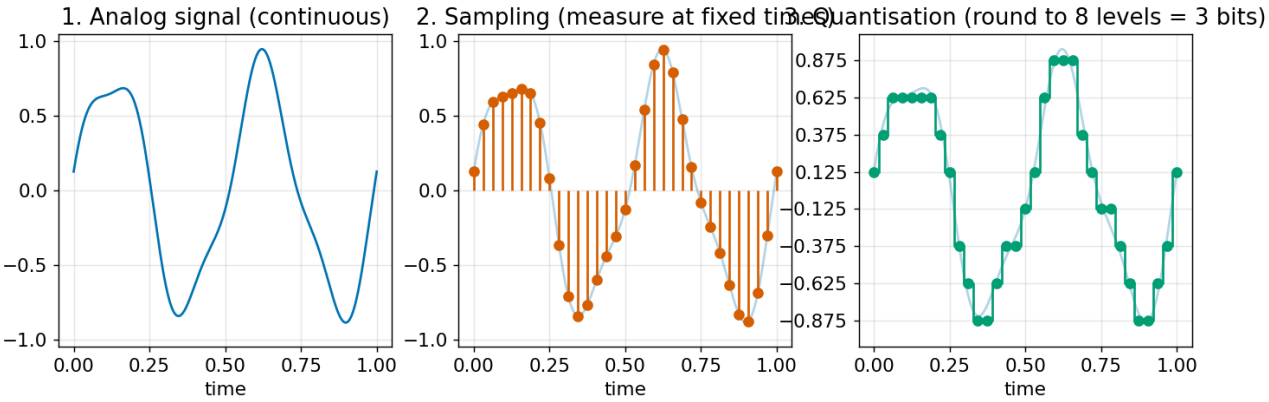


Figure 1. The two jobs of an ADC: sampling, then rounding to a fixed set of levels. Here 3 bits give 8 levels.

The number of **bits** sets how many levels exist: b bits give 2^b levels. Think of a ruler. A 12-bit ADC is a ruler with 4096 marks. A 2-bit ADC has 4 marks. A **1-bit ADC** has only 2: it tells you whether the value is positive or negative, and nothing else.

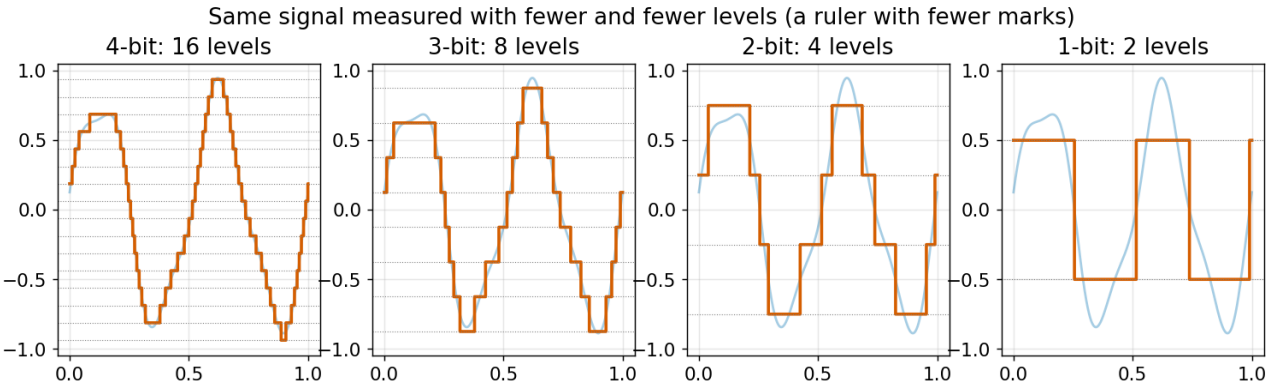


Figure 2. The same signal measured with 4, 3, 2 and 1 bits. With 1 bit only the sign of the signal survives.

Key idea. Fewer bits means coarser numbers. At 1 bit the size of the signal is lost completely; only its sign is kept.

2. How the output is calculated (worked example)

The project uses a standard uniform quantiser. For each scene the ADC range is set to the largest sample, A (this is called peak automatic gain control, AGC). Then:

- step size: D = 2A / 2^b
- output for an input x: D x (floor(x / D) + 0.5), kept inside the range -A to +A
- the real part (I) and the imaginary part (Q) of every sample are quantised separately

Worked example with A = 1. A complex sample is **0.60 - 0.30j**.

	step D	allowed levels	real part 0.60 becomes	imaginary part -0.30 becomes	output
3-bit	0.25	-0.875 ... +0.875 (8 levels)	0.625	-0.375	0.625 - 0.375j
2-bit	0.5	-0.75, -0.25, +0.25, +0.75	0.75	-0.25	0.75 - 0.25j
1-bit	1	-0.5, +0.5	+0.5	-0.5	0.5 - 0.5j

At 1 bit, the samples 0.60 - 0.30j and 0.01 - 0.99j give the **same** output, 0.5 - 0.5j. This is the information that is lost.

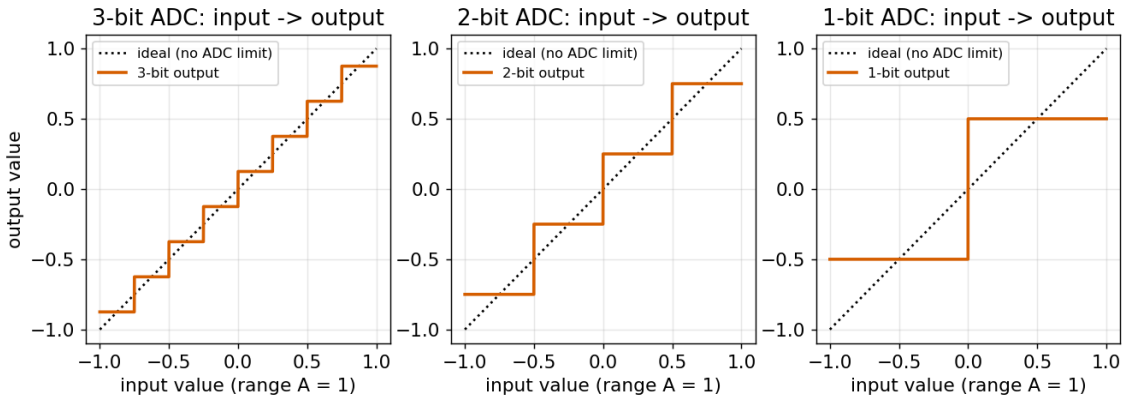


Figure 3. Input-to-output curves. The dotted line is a perfect ADC. The staircase is what a 3-, 2- or 1-bit ADC returns.

3. What the ADC does to the beam measurements

In this project the receiver records 256 complex numbers per scene: one for each pair of a transmit beam and a receive beam (a 16 x 16 grid called Y). Figure 4 shows these 256 numbers as points. After a 1-bit ADC every point lands on one of only 4 values.

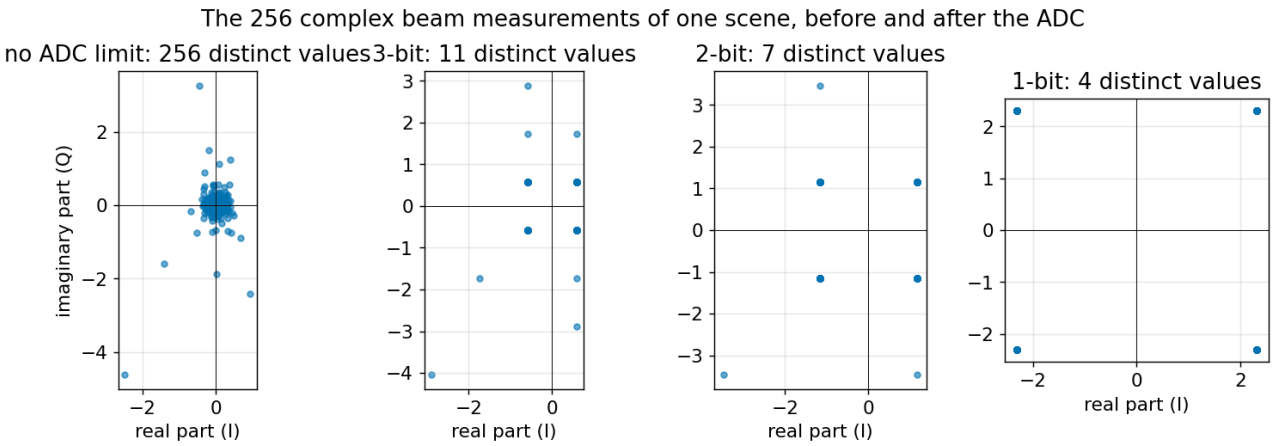


Figure 4. The 256 beam measurements of one simulated scene as points (real part against imaginary part), before and after the ADC.

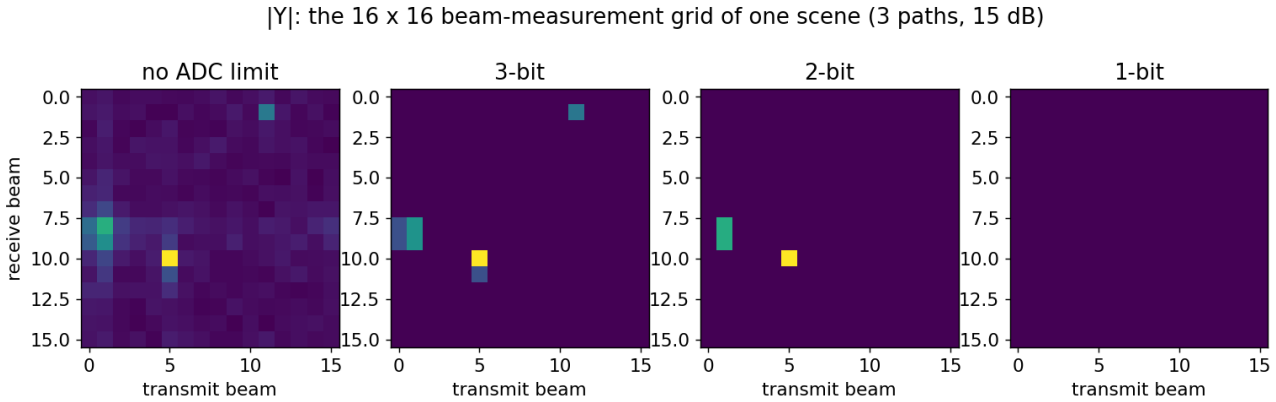


Figure 5. The same scene as a 16 x 16 picture of magnitudes. Bright cells are the beam pairs that point at a signal path. At 1 bit all magnitudes become equal, so the picture is flat: the information is only in the signs.

With peak AGC the range is set by the strongest beam. Most of the other 255 samples are much smaller than one step, so even at 3 or 4 bits most samples keep only their sign.

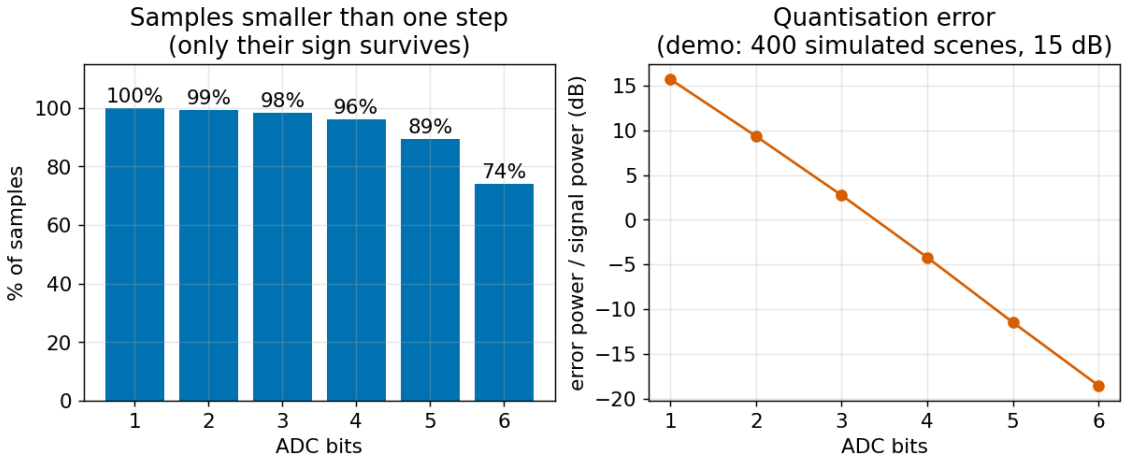


Figure 6. Left: share of samples smaller than one quantisation step. Right: quantisation error relative to the signal. Both are demos on 400 simulated scenes at 15 dB. The error at 1 and 2 bits is larger than the signal because every sample is given the same large size; the decoders ignore the overall scale, so the pattern of signs is what matters.

Key idea. With peak AGC, weak paths behave almost as if the ADC had 1 bit, even when it has 2 or 3. The study notes measured a median loss for the weak path of about 2 dB at 1 to 3 bits.

4. Why low-bit ADCs are used at all

What the papers say:

- **Walden, IEEE JSAC 1999** (the classic ADC survey): for the converters surveyed, resolution falls by about 1 bit for every doubling of the sampling rate. Fast converters are therefore coarse or power-hungry.
- **Mo and Heath, IEEE TSP 2015**: states that ADC power grows exponentially with resolution (citing Walden), and shows that a 1-bit receiver loses at least 10 log10(pi/2) = 1.96 dB of power at low SNR. This is a capacity result, not an angle-estimation result.
- **Li et al., IEEE TSP 2017**: one-bit ADCs need no automatic gain control and no linear amplifiers.
- **Mo, Schniter and Heath, IEEE TSP 2018**: few-bit ADCs "perform comparably to infinite-bit ADCs at low SNR" for channel estimation (measured by channel error, not by angle accuracy).

Be careful. Walden does not state a power law in bits; the "exponential power" statement in later papers is a count of comparators (2^b - 1). This project did not measure ADC power. Use the power argument only as motivation.

5. The theory: what quantisation does to a signal

5.1 Bussgang view (Li et al. 2017)

For a Gaussian input, a 1-bit ADC can be written as a **scaled copy of the input plus a distortion**. The scale is sqrt(2/pi) = 0.798, and the distortion is uncorrelated with the input.

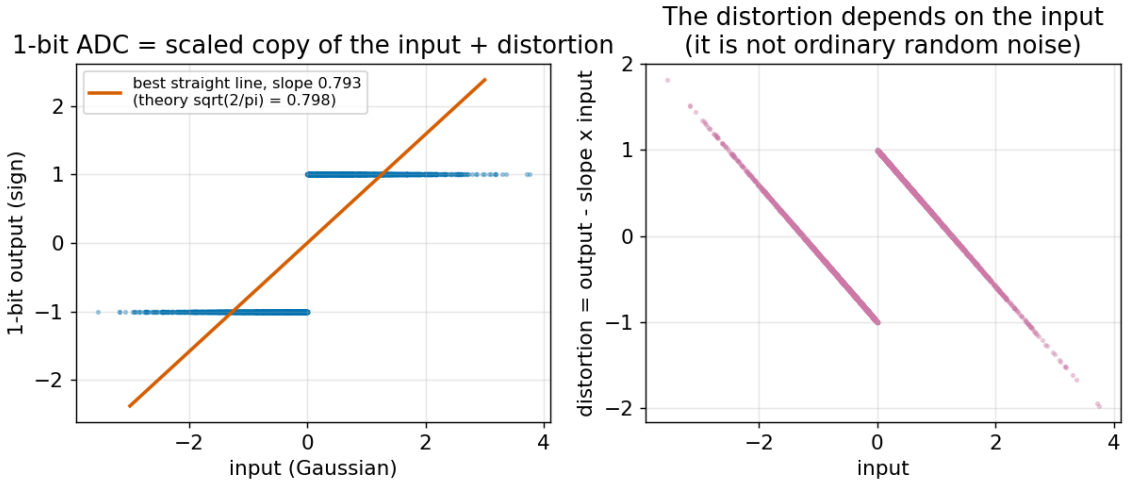


Figure 7. Demo with 4000 Gaussian samples. Left: the best straight line through the 1-bit output has slope 0.79, matching sqrt(2/pi). Right: the distortion is not ordinary noise; its size depends on the input.

This idea leads to a simple fix: undo the scaling with a linear filter (called BLMMSE), then use an ordinary estimator. **In this project that fix does not work.** Bussgang filtering followed by NOMP was worse than plain NOMP in all four cases tested (for example 0.053 against 0.161 at 1 bit, 0 dB). The beam measurements are not Gaussian: a few beams are very strong and most are weak.

5.2 Exact likelihood view (Mo et al. 2018; GNOMP, Zhu et al. 2026)

The better idea is to ask a different question. Do not ask "which angles make my model closest to the ADC output?". Ask: **"if the angles were these, how probable is it that the ADC would report exactly the cells I saw?"** For each sample, the model predicts a value; noise spreads it into a bell curve; the probability of the observed cell is the area of the bell curve inside that cell.

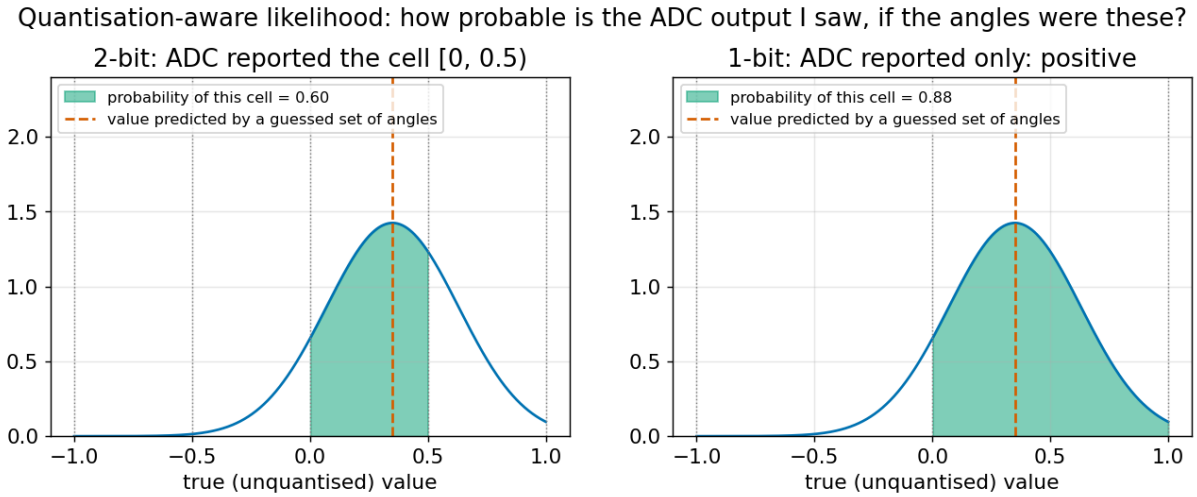


Figure 8. The cell probability for one sample. Multiplying these probabilities over all 512 real numbers (256 samples, I and Q) gives the quantisation-aware likelihood. The best angles are the ones that make it largest.

This exact likelihood is **prior art**. It is used with message passing in Mo, Schniter and Heath (2018) and with NOMP-style search in GNOMP (Zhu et al., IEEE TAES 2026). The project's "quantiser ML" baseline uses the same likelihood with a different search (13 starting points plus refinement).

Key idea. Quantisation is a known, exact rule. A method that models it correctly does much better than one that treats the ADC output as if it were the true signal.

6. Why NOMP struggles with 1-bit data

NOMP (Mamandipoor, Ramasamy and Madhow, IEEE TSP 2016) finds the strongest peak in the angle spectrum, polishes it with Newton steps, removes it and repeats. It assumes the data are the true signal plus ordinary Gaussian noise. Figure 9 shows the spectrum NOMP searches for the same scene.

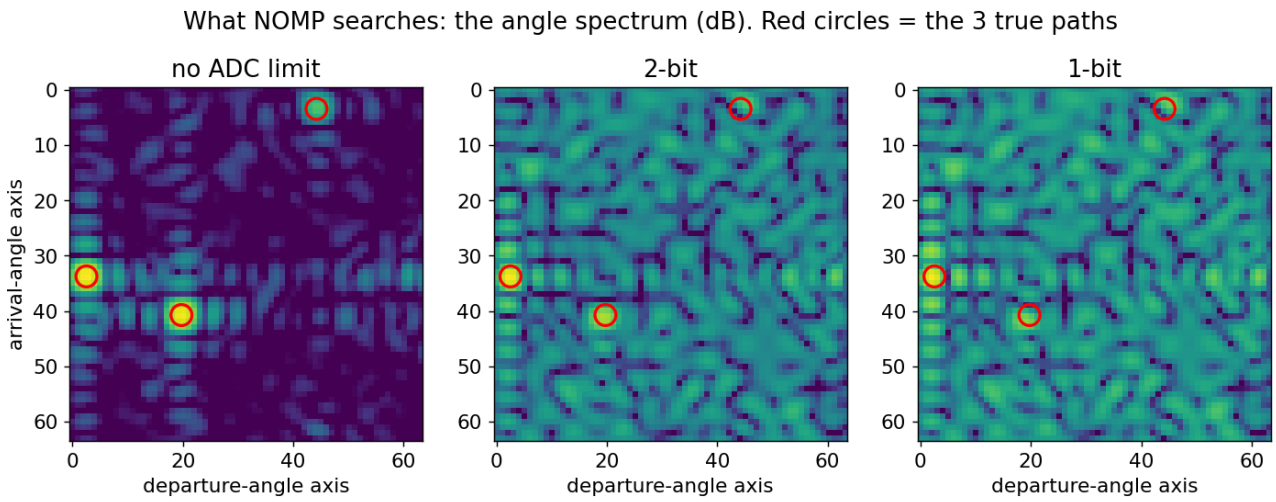


Figure 9. Angle spectrum of one scene. Without an ADC limit the three paths stand out clearly. After a 2-bit or 1-bit ADC the weakest path is close to the level of the quantisation clutter.

Two things go wrong:

- **Clutter.** Quantisation distortion is spread over the whole spectrum, so weak paths are hard to tell from false peaks.
- **A biased fit.** NOMP's refinement minimises squared error to the ADC output. For quantised data the best squared-error fit is not at the true angles. In a project check at 1 bit, 15 dB, starting NOMP's refinement from the true angles (rounded to its search grid) lowered the score from 0.658 to 0.430.

NOMP on ideal hardware scores Pd 0.758. On 1-bit data at 15 dB it scores 0.191.

7. The project's approach and what the results show

The hybrid: the network finds where to look, the physics-based step makes it precise

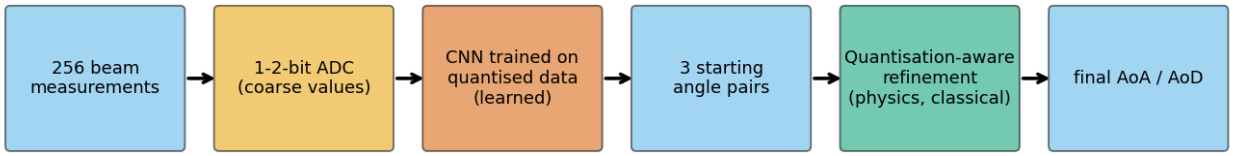


Figure 10. The hybrid method. The CNN is trained on quantised data and gives starting angles. The refinement uses the exact quantisation-aware likelihood of Section 5.2.

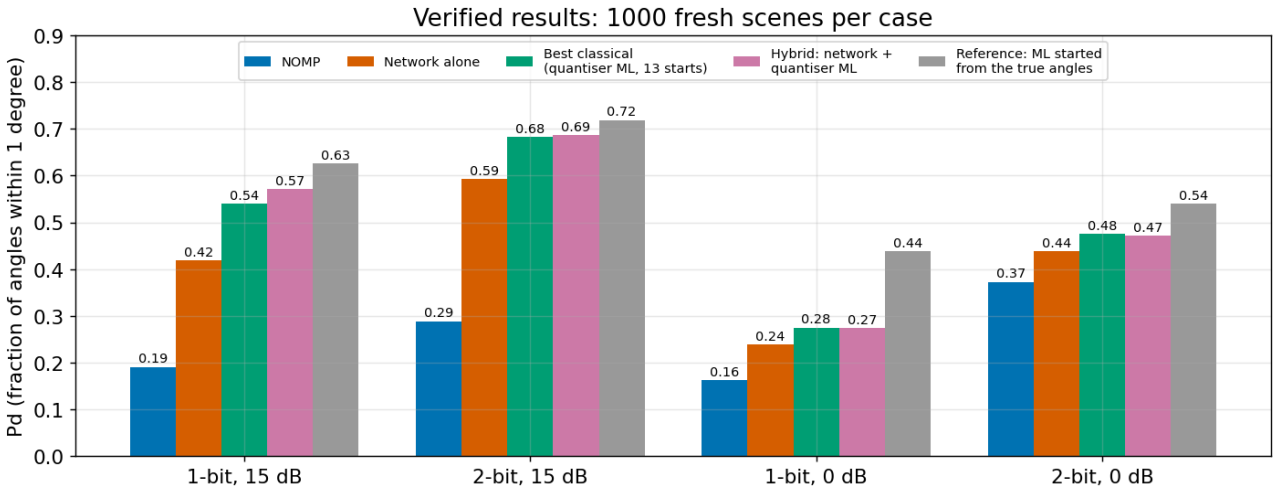


Figure 11. Verified results. Pd is the fraction of angles estimated within 1 degree.

Question	Answer from the verified results
Is the network better than NOMP on 1-2-bit data?	Yes, clearly. At 1 bit, 15 dB: 0.419 against 0.191. At 2 bits, 15 dB: 0.593 against 0.289. The time per scene is about the same (6.2 ms against 5.8 ms).
Is the network alone better than the best classical method?	No. Quantisation-aware maximum likelihood with 13 starts scores 0.540 and 0.682 in those cases. It is slower: 482 ms per scene.
Is the hybrid (network + quantisation-aware refinement) better than the best classical method?	At 1 bit, 15 dB: yes, 0.572 against 0.540 (+0.032), confirmed again on new data (+0.028 and +0.036). In the other three cases it ties.
What is not yet tested?	GNOMP extended to two angles, and starting points from a training-free method (EM-VAMP). Both are needed before claiming the network is necessary.

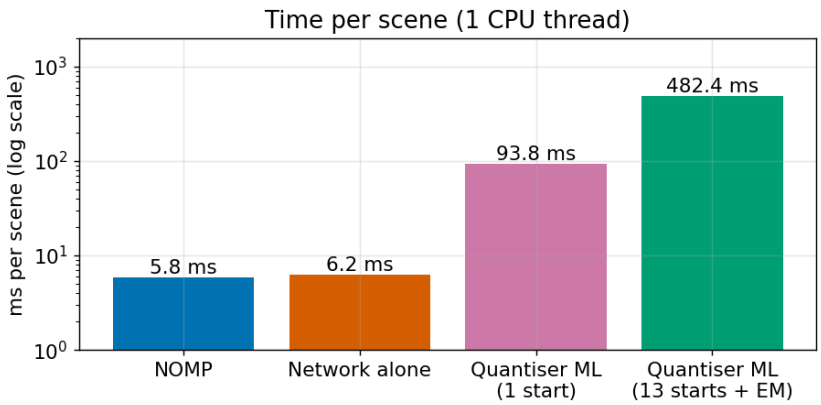


Figure 12. Time per scene on one CPU thread. The network alone reaches 78 to 92% of the best classical accuracy at about 1/80 of its time.

Key idea. Honest summary: the network is far better than NOMP on 1-2-bit data. It is not better than a classical method that models the quantiser. The combination of the two is the best result so far, by a small margin, at 1 bit and 15 dB.

8. What the nearest research papers say (short notes)

Paper	What it does	What to remember
Mamandipoor, Ramasamy, Madhow. "Newtonized Orthogonal Matching Pursuit". IEEE TSP 64(19), 2016	Grid search, Newton refinement, cyclic re-polish. One-dimensional, unquantised, Gaussian noise.	The classical decoder used here (extended to two angles). Its noise model is wrong for coarse ADC data.
Zhu, Zhang, Zhang, Fang, Qu. GNOMP. IEEE TAES 62, 2026	NOMP with the exact cell likelihood for few-bit data. One-dimensional.	The closest classical competitor. Loss of 4.8 / 2.3 / 1 dB at 1 / 2 / 3 bits in one of its scenarios. A two-angle version has not yet been validated in this project.
Mo, Schniter, Heath. IEEE TSP 66(5), 2018	Few-bit mmWave channel estimation with the exact likelihood and message passing (EM-GAMP / EM-VAMP).	Reports channel error, not angle accuracy. Noise level assumed known.
Li, Tao, Seco-Granados, Mezghani, Swindlehurst, Liu. IEEE TSP 65(15), 2017	Bussgang decomposition and BLMMSE channel estimator for 1-bit massive MIMO.	The scaled-copy-plus-distortion view. In this project the linear fix made NOMP worse.
Dreifuerst, Heath. SignalNet. IEEE TSP 70, 2022	Neural network for sinusoids from 1- and 3-bit samples.	Warns that at 1 bit a network may learn the typical answer instead of the data. Control tests are needed.
Hu, Sun, Zhang. IEEE JSTSP 19(6), 2025	Unrolled network for 1-bit single-snapshot direction finding.	The learned version matches its classical parent at 0-15 dB.
Khobahi, Shlezinger, Soltanalian, Eldar. LoRD-Net. IEEE TSP 69, 2021	Unfolded 1-bit symbol detector with learned preconditioners.	Its reported gain over the classical method looks partly due to a weak baseline (project re-implementation).
Walden. IEEE JSAC 17(4), 1999; Mo and Heath. IEEE TSP 63(20), 2015	ADC survey; capacity of 1-bit MIMO.	Motivation for low-bit ADCs; the 1.96 dB minimum loss.

Status of these notes. Each paper was read in full by the project's study agents, mostly from the arXiv version; for several, the published journal text was not separately checked. Numbers read from plots are approximate. Check the exact wording and page numbers against the PDFs before quoting. The full notes are in *paper/evidence/adc_program/papers/* and *synthesis/TAKEAWAYS.md*.

9. Limits of the current results

- One ADC model only: uniform levels with peak AGC, for both training and testing.
- The number of paths (3) is given to every method.
- All results are simulation; no measured data.
- No gain at 0 dB, and no advantage at 3-4 bits, where the classical decoder is better.
- Two important baselines are still missing: GNOMP for two angles and EM-VAMP starting points.

10. Glossary

Term	Plain meaning
ADC	Analog-to-digital converter: turns a voltage into a number.
Bit	b bits give 2^b possible output levels.
Quantisation	Rounding each sample to the nearest allowed level.
AGC (peak)	Setting the ADC range to the largest sample of the scene.
I and Q	The real and imaginary parts of a complex sample; each has its own ADC.
Beam measurement grid Y	The 16 x 16 table of samples, one per transmit-beam / receive-beam pair.
NOMP	Classical estimator: find a peak, refine it with Newton steps, remove it, repeat.
Likelihood	How probable the observed data are for a guessed set of angles.
Quantiser ML	Choosing the angles that make the observed ADC cells most probable.
Bussgang decomposition	Writing the ADC output as a scaled copy of the input plus a distortion.
Pd	Fraction of angles estimated within 1 degree of the truth.
SNR	Signal-to-noise ratio in dB; higher is cleaner.